

Anthony Marchello

Milwaukee, WI

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SUMMARY

Software Engineering B.S. candidate with a Data Science minor at the Milwaukee School of Engineering. Experienced in team-based projects involving embedded systems, front-end and back-end development, DevOps, and data model creation. Worked with Scot Forge on a year-long academic internship, developing a web application with Python and JavaScript in a team environment. Contributing to a Senior Design team developing a digital audio workstation (DAW) using the JUCE C++ framework and Meta's MMS Text-to-Speech (TTS) model.

EDUCATION

Software Engineering B.S. | Milwaukee School of Engineering | GPA: 3.28 | Expected Graduation: May 2026

Data Science Minor | Milwaukee School of Engineering | Expected Completion: May 2026

Google Cybersecurity Professional Certificate | Coursera | Expected Completion: TBD in 2026

INTERNSHIP EXPERIENCE

Software Developer Intern | Scot Forge | Remote (Headquartered: Spring Grove, IL) | September 2024 - May 2025

Project Objective: Worked on a web app Part Creation Tool (PCT) to help show customers what parts they will be buying.

- Helped fix rendering issues (similar to SolidWorks, dimensions are inputted to create a part).
- Improved backend (updated APIs to MongoDB) and frontend (React JS).
- Assisted with deploying the web app on a server using Docker.
- Worked in an Agile environment with our manager throughout the year, delivering high-priority PBIs.

TECHNICAL SKILLS

Programming Languages: C, C++, C#, Java, Python, R, Rust, JavaScript, HTML, CSS, x86 Assembly, SQL

Web Development: React.js, Express.js, Bootstrap, GitHub Pages

Databases: MySQL, MongoDB, Firebase

Cloud & DevOps: AWS, Docker, Linux Command Line, GitLab

Data Science & Machine Learning: Pandas, NumPy, TensorFlow, Keras

Software Engineering: Algorithms (Greedy, Divide-and-Conquer, etc.), Data Structures (Self-Balancing Trees, Heaps, etc.), Software Design patterns (Singleton, Observer, etc.), Testing Theory (Black Box, White Box, etc.), Agile Software Development (Scrum, Sprints, Daily Standups, etc.)

Other Knowledge: Poetry, Mock Objects, Raspberry Pi, Robot Control

PROJECT EXPERIENCE

Scot Forge Web Application

- Collaborated on a year-long remote intern team with a metal forging company.
- Assisted with back-end and front-end development.
- Learned to deploy applications using Docker and manage Python projects with Poetry.
- Explored techniques for reducing technical debt in code.

React.js Personal Website

- Designed a user-friendly interface using React.js and JSX.
- Applied modular JavaScript to enhance maintainability.
- Deployed the website via GitHub Pages.

ROM Hack of Pokémon Sapphire

- Learned to use the game development tool Porymap to edit the 2D world map.
- Added interactive elements to the world map.
- Used the Linux command line (via Debian) to rebuild the project executable.
- Leveraged public repositories and tools to support project development.

JUCE DAW with Nvidia's Fugatto model

- Year-long senior design group project using the JUCE C++ framework.

- Planning a collaborative music composition feature like Google Docs or Microsoft Word.
- Will use Nvidia's Fugatto model along with Dolby Atmos technology.

AWS Prime Number Determinator Application

- Can determine if a large integer is a prime number.
- Utilized lambdas to enhance functionality.
- Employed queues for efficient data handling.

Python Classification of Japanese Characters

- Utilized KMNIST to train a model for identifying Japanese characters.
- Prepared data for a deep learning model.
- Compared the performance of CNN and DNN models.

Module JavaScript (mjs) Text File Manager Web Application

- Utilized Module JavaScript (mjs) and JavaScript to run a server on my local computer.
- Enabled users to post text on a web page.
- Allowed users to edit the text after posting.

C++ Clone of Hunt the Wumpus Game

- Created the clone with a classmate.
- Utilized separate .cpp and .h files for organization.
- Implemented multiple classes to structure the game.

React JS Culver's Menu Web Application

- Utilized Bootstrap for design.
- Enabled users to filter the menu by allergen or item name.

ACHIEVEMENTS

- Earned Dean's List status during Fall Semester of Junior and Senior years.
- Earned Honors status during Fall Semester of Sophomore year.

CO-CURRICULAR INVOLVEMENT | COMMUNITY SERVICE

Concert Band Member | MSOE Concert Band | December 2022 - Present

Church Choir Member | St. Matthew's Church | August 2017 - Present

Video Game Dev. Member | MSOE Video Game Development Club | September 2023 - Present

Circle K Member | MSOE Circle K | September 2023 - Present

Sexuality and Gender Alliance (SAGA) Member | MSOE Sexuality and Gender Alliance | September 2023 - Present

Salvation Army Bell Ringing Member | Salvation Army | December 2018 - Present

WORK EXPERIENCE

Deli Associate | Sendik's Food Market, West Bend | March 2022 - Present | 38 to 40 hours per week (during breaks)

Front Desk Worker/Student Assistant | Milwaukee School of Engineering | March 2023 - May 2024

Full-Service Gas Station Attendant | Herrige's Oil/Mike's Auto | November 2018 - December 2021

INTERESTS | HOBBIES | OTHER

I enjoy reading, playing video games, meditating, walking my dog, programming my ROM hack, hunting pheasants, freshwater fishing, following football, watching horror movies with friends, and traveling with my family. I also have a deep passion for open-wheel racing, having developed a slogan for the 2015 Milwaukee IndyCar Race and job-shadowed an IndyCar team.